

dompatent von Kreisler | P.O. Box 10 22 41 | 50462 Köln/Germany

European Patent Office
Bob-van-Bentham-Platz 1
80469 Munich

Contact:
Dr. Christoph Schreiber

cshreiber@dompatent.de

Tel.:
+49 221 91652-228

Our Ref.:
172985EP CS/vg

6 August 2020

European Patent Application 16802532.8
USE OF PTERIDINONE DERIVATIVE SERVING AS EGFR
INHIBITOR
East China University of Science and Technology

Responsive to the communication dated 19 March 2020:

A new set of 8 claims is submitted in mark-up and clean copy which should be the basis for the further prosecution.

In claim 1, the definition of R1 was limited to "substituted or unsubstituted phenyl, optionally substituted C₃-C₈ cycloalkyl".

The dependent claims were adapted.

In Claim 3 some additional compounds were added, disclosed in the application. These compounds are identified by the number used in the specification.

Köln

Patentanwälte | Patent Attorneys

Jochen Hilleringmann, Dipl.-Ing.
Dr. Hans-Peter Jönsson, Dipl.-Chem.
Dr. Thomas Weber, Dipl.-Chem.
Dr. Jörg Helbing, Dipl.-Chem.
Alexander von Kirschbaum, Dipl.-Ing.
Dr. Christoph Schreiber, Dipl.-Chem.
Dr. Thomas Escher, Dipl.-Chem.
Constantin Rühlend, Dipl.-Ing.
Florian Krüger, Dipl.-Ing.
Kerstin Renner, Dipl.-Ing.
Andreas Gagidis, Dipl.-Ing. (BA)
Dr. Eva Bock, Dipl.-Chem., LL.M.
Dr. Marius Vieweg, Dipl.-Phys.
Dr. Anke Krebs, Dipl.-Chem., LL.M.
Dr. Alan Stewart, B.Sc. (Hons) Chem.
Dr. Alexander Schuld, Dipl.-Ing.
Dr. Reza Ghadiri, Dipl.-Ing.

Rechtsanwälte | Attorneys at Law

Dr. Felix Hauck, LL.M.
Dr. Alexander Theis, LL.M.
WanZe Zhang, LL.M.

Of Counsel

Patentanwälte | Patent Attorneys

Georg Dallmeyer, Dipl.-Ing.
Dr. Hans-Wilhelm Meyers, Dipl.-Chem.

On the other hand, some compounds were deleted which no longer belong to the invention. This also applies to claim 5.

Based on these amendments, compounds 19, 20, 31 and 32 of D4 are no longer covered.

Regrading inventive step, the compounds of the present invention differ significantly from those of D4.

Regarding compound 19 of D4, this compound is not a good EGFR inhibiting compound. Its IC₅₀ value of inhibitory activity on T790M/L858R-EGFR is 738 nM, and its IC₅₀ value of inhibitory activity on the proliferation of H1975 (which is a non-small cell lung cancer cell, EGFR T790M/L858R) is only 8.87 μ M. Moreover, this compound does not exhibit good selectivity between WT EGFR and T790M/L858R-EGFR. For example, the ratio of the IC₅₀ value of inhibitory activity on EGFR of wild-type to that on mutant EGFR is merely 1.70.

Regarding compound 20 of D4, this compound does not exhibit better inhibitory activity on the proliferation of H1975 than the control compound, and it is well-known for a skilled person that the inhibitory activity on proliferation of cells is more important than that on kinase. Moreover, this compound does not exhibit good selectivity between WT EGFR and T790M/L858R-EGFR. For example, the ratio of the IC₅₀ value of inhibitory activity on EGFR of wild-type to that on mutant EGFR is merely 1.78.

Regarding compound 031 of D4, this compound does not have better EGFR inhibitory activity than the control compound WZ4002 and does not exhibit good selectivity between WT EGFR and T790M/L858R- EGFR. For example, the ratio of

the IC₅₀ value of inhibitory activity on EGFR of wild-type to that on mutant EGFR is merely 1.87.

Regarding compound 032 of D4, this compound exhibits comparable inhibitory activity on the proliferation of H1975 to the control compound WZ4002, and the ratio of the IC₅₀ value of inhibitory activity on EGFR of wild-type to that on mutant EGFR is 3.14.

Compared with the compounds of D4, the compounds of the invention exhibit a significantly better activity in at least one of the following aspects:

1) excellent EGFR inhibitory activity. For example, the IC₅₀ values of inhibitory activity on proliferation of cells (H1975) of compounds 4, 5, 12 and 42 are 0.009, 0.006, 0.006 and 0.003 μ M, respectively;

2) excellent selectivity between WT EGFR and T790M/L858R- EGFR, or between cells with WT EGFR and cells with T790M/L858R- EGFR. For example, the IC₅₀ values of inhibitory activity on proliferation of cells (H1975) of compound 25 is 0.016 μ M, however, the ratio of the IC₅₀ value of inhibitory activity on EGFR of wild-type to that on mutant EGFR is **43.47** and the ratio of the IC₅₀ value of inhibitory activity on proliferation of A431 cells (wild-type EGFR) to that on H1975 cells (mutant EGFR) is **109.38**. For another example, the IC₅₀ values of inhibitory activity on proliferation of cells (H1975) of compound 53-2(S) is 0.062 μ M, however, this compound exhibits excellent selectivity between cells with WT EGFR (> 10 μ M) and cells with T790M/L858R- EGFR (0.062 μ M).

3) significantly different structures. For example, R1 of formula I or R1 of the invention, for example compounds 43, 47, 48, 55-57, is a substituted phenyl or

cycloalkyl, however, in the compounds of D4, the substituents corresponding to R1 of the compounds as claimed in the amended claims are H or methyl.

It is kindly requested to confirm allowability of the set of claims. An adapted description will subsequently be submitted.

Dr. Christoph Schreiber
European Patent Attorney

Enclosures: amended claims (mark-up)
amended claims (clean copy)